ELEC205 Assignment 2

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Question 1

The Moore state diagram for the dual sequence detector is illustrated in Figure 1. The design utilizes five states (S0 through S1) to monitor the input X and detect the target sequences 0 -> 1 and 0 -> 1 -> 1 . To satisfy the non-overlapping requirement, the circuit transitions from completion states (S3, S4) directly back to S1 if X = 0 or to S0 if X = 1, ensuring immediate re-detection without pattern reuse.

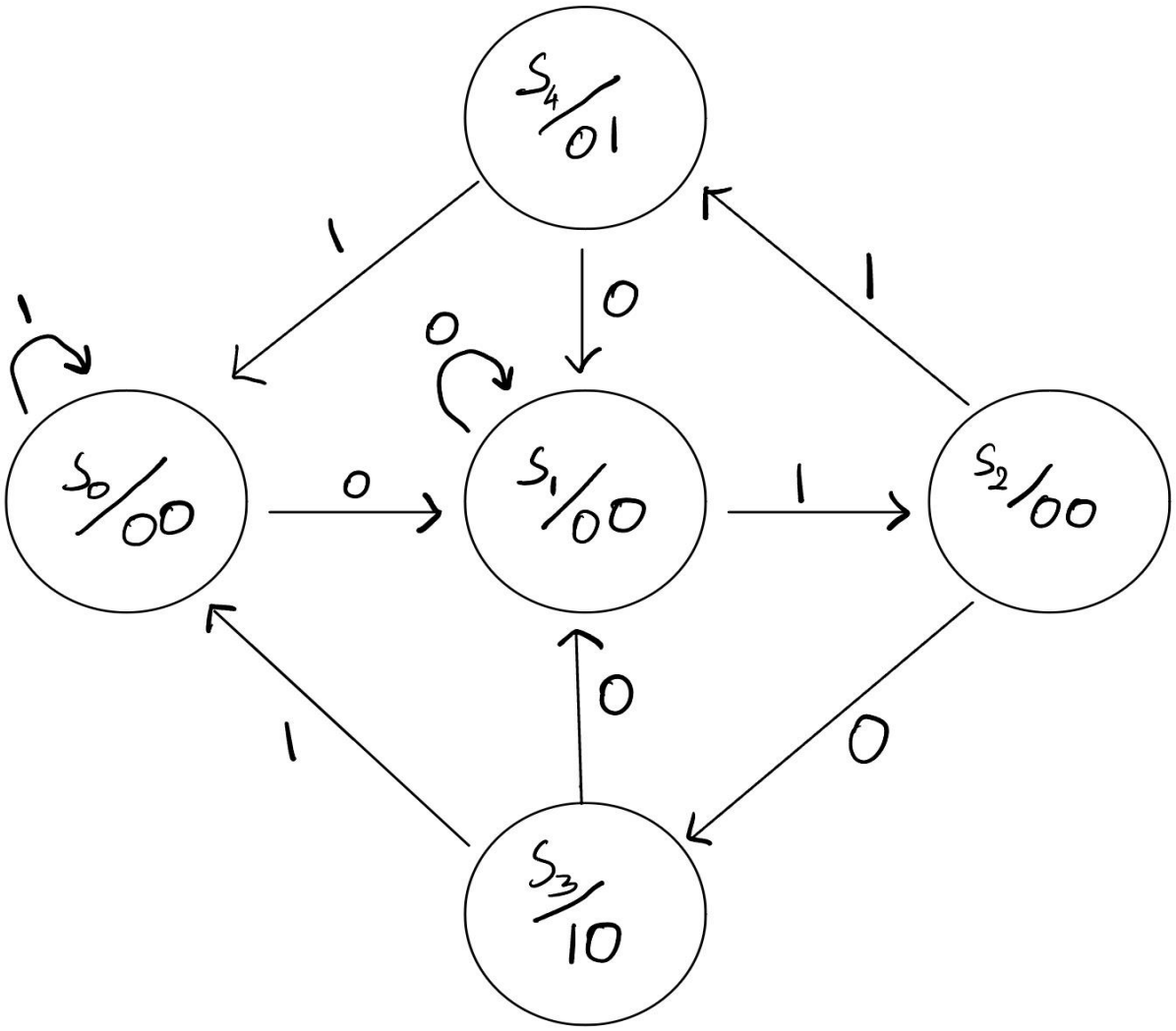


Figure 1

Question 2

In this step, binary codes are assigned to each state to represent them in the digital circuit. Since there are five distinct states, three bits (Q2, Q1, Q0) are required. The chosen encoding for each state is summarized in the table below.

State	Binary Code
S0	000
S1	001
S2	010
S3	011
S4	100

Table 1

Question 3

The transition logic and output values are detailed in the table below. It defines the next state (D2, D1, D0) and outputs (Z1, Z0) for every possible combination of the current state and input X. Unused binary codes are marked as "don't care" (X) to simplify the circuit design.

Present State (Q2Q1Q0)	**Input (X)**	**Next State (D2D1D0)**	**Output (Z1Z0)**
S0 (000)	0	001	00
S0 (000)	1	000	00
S1 (001)	0	001	00
S1 (001)	1	010	00
S2 (010)	0	011	00
S2 (010)	1	100	00
S3 (011)	0	001	10
S3 (011)	1	000	10
S4 (100)	0	001	01
S4 (100)	1	000	01
X (101)	0	XXX	XX
X (101)	1	XXX	XX
X (110)	0	XXX	XX
X (110)	1	XXX	XX
X (111)	0	XXX	XX
X (111)	1	XXX	XX

Table 2

Question 4

In this section, Karnaugh Maps (K-maps) are utilized to derive the simplified Boolean expressions for each next-state variable (D2,D1,D0) and system outputs (Z1,Z0). The "don't

care" conditions originating from unused states are leveraged to achieve maximum simplification.

As per the design constraints, all expressions are converted into a format suitable for NAND-only implementation. This is achieved by applying De Morgan's laws and the principle of double negation, where an AND-OR structure is equivalent to a NAND-NAND hierarchy.

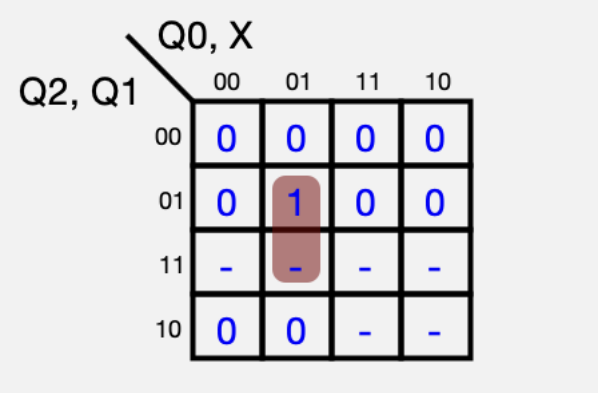


Figure 2

D2 = Q1.Q0'.X = (Q1.Q0'.X)"

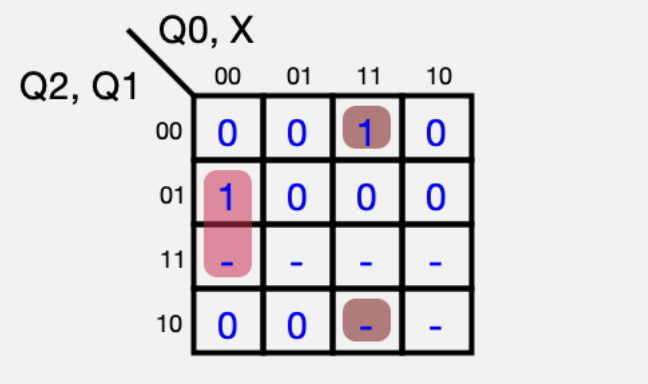


Figure 3

D1 = (Q1'.Q0.X) + (Q1.Q0'.X') = ((Q1'.Q0.X)' . (Q1.Q0'.X')')'

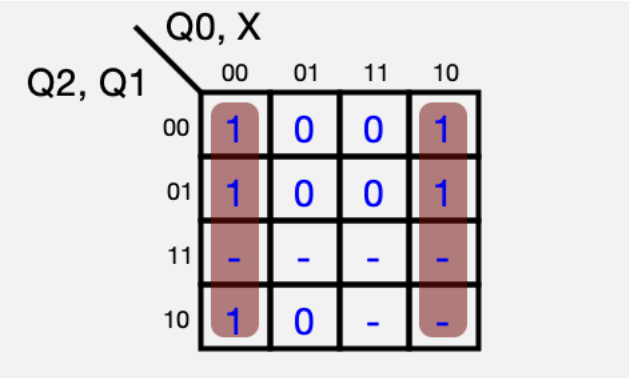


Figure 4

D0 = X'

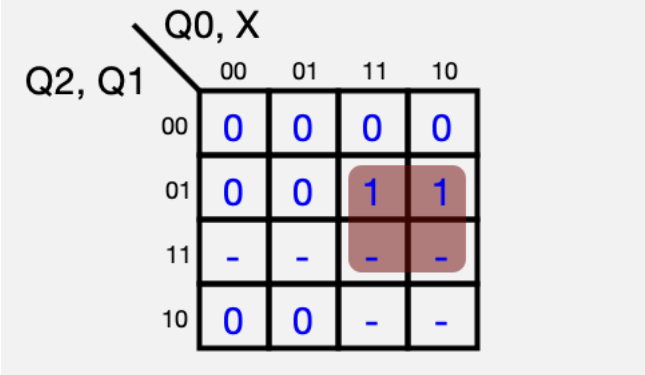


Figure 5

Z1 = Q1.Q0 = (Q1.Q0)"

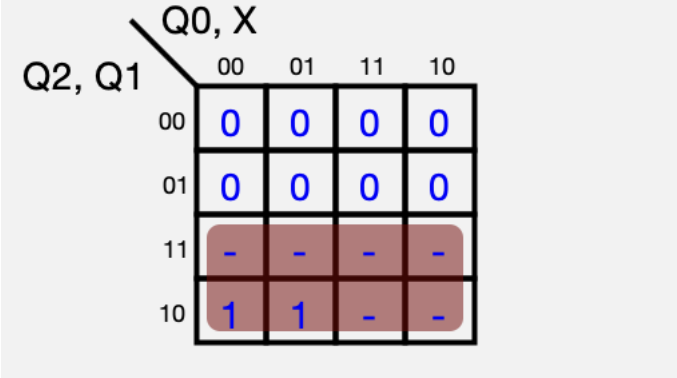


Figure 6

Z0 = Q2

Question 5

The complete logic circuit is implemented in Logisim using three D flip-flops and NAND gates. The state variables Q2, Q1, Q0 and their complements are routed to the combinational logic block to generate the next-state inputs (D2, D1, D0). All AND-OR logic structures are

realized using a NAND-only hierarchy to comply with the design constraints. A common clock signal synchronizes the state transitions.

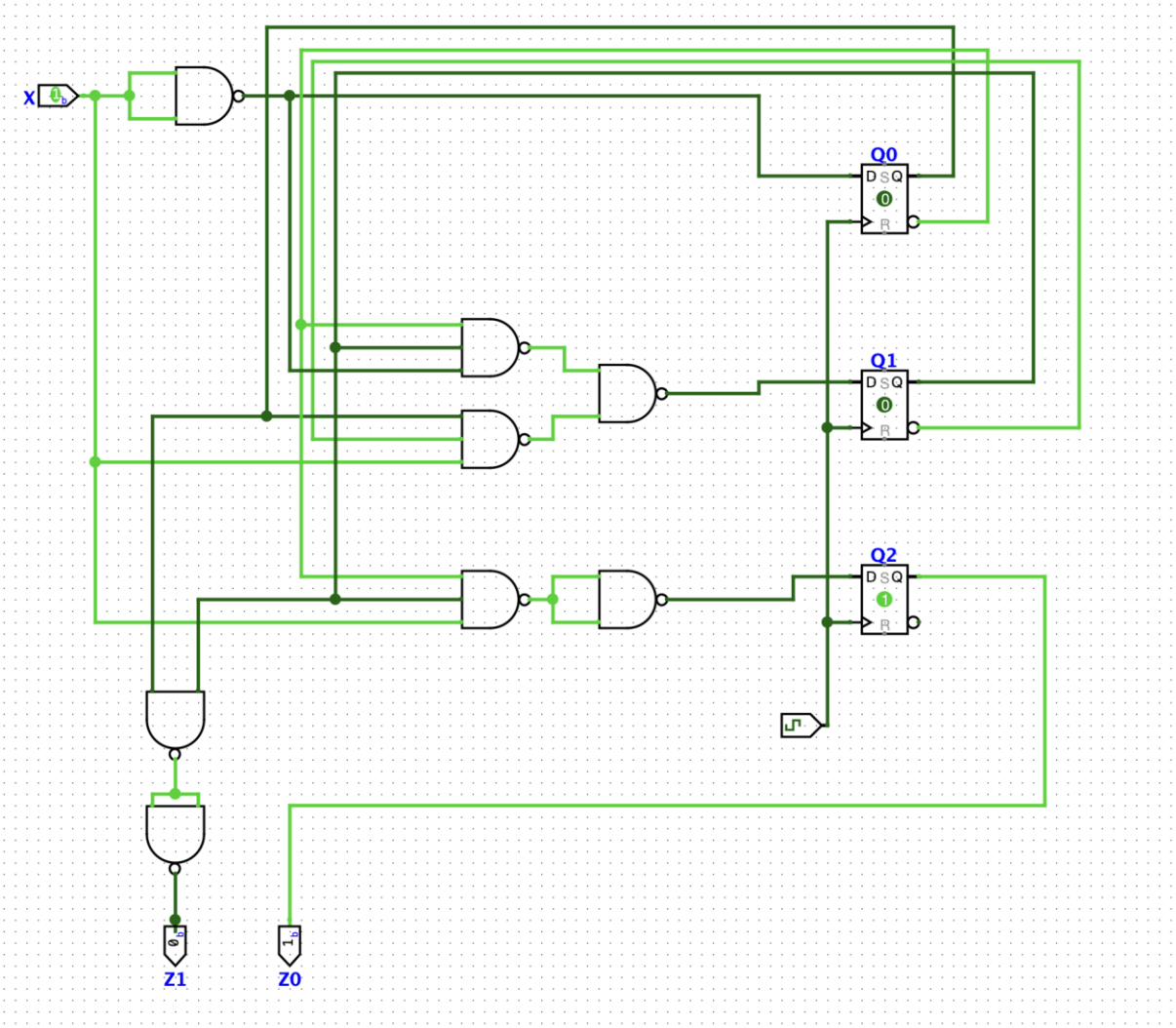


Figure 7

Question 6

The circuit's behavior was validated by manually toggling the input X and observing the state transitions. When the input sequence 0-1-0 is provided, the output Z1 pulses high for exactly one clock cycle. Similarly, the sequence 0-1-1 triggers the Z0 output. The transition from completion states back to S1 or S0 was also verified, confirming that the circuit correctly follows the no-overlap and immediate restart rules.

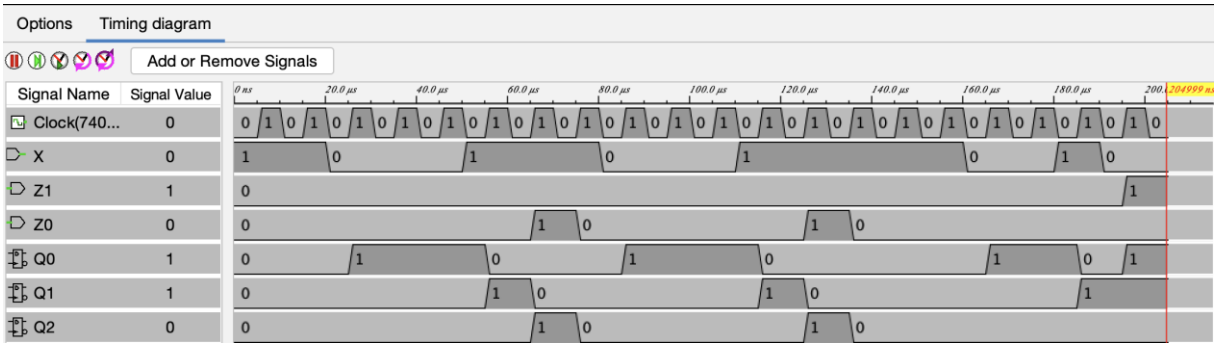

Options
Timing diagram
Add or Remove Signals
Signal Name
Signal Value
Clock(740...
0
X
0
Z1
1
Z0
0
Q0
1
Q1
1
Q2
0
0 ns
20.0 µs
40.0 µs
60.0 µs
80.0 µs
100.0 µs
120.0 µs
140.0 µs
160.0 µs
180.0 µs

Figure 8